

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12409317
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Govari; Assaf et al.

Balloon catheter with ultrasonic transducers

Abstract

A balloon catheter that has one or more ultrasonic transducers to map a body cavity and to measure thicknesses of walls in the body cavity during an ablation procedure. The balloon of the catheter is deployable through a lumen into the body cavity with a flexible printed circuit board that has a first side attached to the exterior wall of the inflatable balloon and a second side opposite the first side. There is also provided an ultrasonic transducer mounted on the first side of the flexible printed circuit board and encapsulated between the exterior wall of the balloon and the flexible printed circuit board. A method of making or using the balloon catheter is also provided.

Inventors:	Govari; Assaf (Haifa, IL), Beeckler; Christopher Thomas (Brea, CA)
Applicant:	BIOSENSE WEBSTER (ISRAEL) LTD. (Yokneam, IL)
Family ID:	61868422
Assignee:	BIOSENSE WEBSTER (ISRAEL) LTD. (Yokneam, IL)
Appl. No.:	15/477171
Filed:	April 03, 2017

Prior Publication Data

Document Identifier	Publication Date
US 20180280658 A1	Oct. 04, 2018

Publication Classification

Int. Cl.: A61M25/00 (20060101); A61B5/00 (20060101); A61B5/06 (20060101); A61B5/287 (20210101); A61B8/00 (20060101); A61B8/08 (20060101); A61B8/12 (20060101); A61M25/10 (20130101); A61N1/04 (20060101); A61N1/05 (20060101); A61B5/107 (20060101); A61B18/14 (20060101)

U.S. Cl.:

CPC **A61N1/05** (20130101); **A61B5/063** (20130101); **A61B5/287** (20210101); **A61B5/6853** (20130101); **A61B8/0883** (20130101); **A61B8/12** (20130101); **A61B8/4254** (20130101); **A61B8/445** (20130101); **A61N1/04** (20130101); A61B5/1076 (20130101); A61B5/6886 (20130101); A61B18/1492 (20130101); A61M2025/1075 (20130101)

Field of Classification Search

CPC: A61B (8/445)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4917097	12/1989	Proudian et al.	N/A	N/A
5391199	12/1994	Ben-Haim	N/A	N/A
5443489	12/1994	Ben-Haim	N/A	N/A
5558091	12/1995	Acker et al.	N/A	N/A
5722972	12/1997	Power et al.	N/A	N/A
5865801	12/1998	Houser	N/A	N/A
5944022	12/1998	Nardella et al.	N/A	N/A
5971955	12/1998	Nap et al.	N/A	N/A
5983126	12/1998	Wittkampf	N/A	N/A
6172499	12/2000	Ashe	N/A	N/A
6177792	12/2000	Govari et al.	N/A	N/A
6456864	12/2001	Swanson et al.	N/A	N/A
6514249	12/2002	Maguire	606/37	A61B 18/00
6690963	12/2003	Ben-Haim et al.	N/A	N/A
6788967	12/2003	Ben-Haim et al.	N/A	N/A
7101368	12/2005	Lafontaine	606/21	A61B 18/02
8819928	12/2013	Nix et al.	N/A	N/A
2004/0249401	12/2003	Rabiner et al.	N/A	N/A
2008/0243214	12/2007	Koblish	600/374	A61B 5/0422
2012/0265188	12/2011	Buchbinder	606/21	A61F 7/123
2014/0276789	12/2013	Dandler et al.	N/A	N/A
2015/0025532	12/2014	Hanson	606/41	A61B 18/1492
2015/0366508	12/2014	Chou	600/374	A61B 8/4477
2016/0051321	12/2015	Salahieh	600/439	A61B 1/00082
2018/0146948	12/2017	Chou	N/A	A61B 8/12

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103784195	12/2013	CN	N/A
WO 88/09150	12/1987	WO	N/A
WO 93/04727	12/1992	WO	N/A
WO 2016/183285	12/2015	WO	N/A

OTHER PUBLICATIONS

European Search Report dated Aug. 7, 2018, EP Application No. 18165387.4. cited by applicant
National Intellectual Property Administration, P.R. China, First Office Action, date of issue Jul. 20, 2022, translated to English, 15 pages. cited by applicant

Primary Examiner: Farag; Amal Aly

Attorney, Agent or Firm: Calderon Safran & Wright P.C.

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates generally to invasive medical probes, and specifically to a balloon catheter comprising one or more ultrasonic transducers.

BACKGROUND OF THE INVENTION

(2) A balloon catheter comprises an inflatable balloon at its distal end that can be inflated and deflated as necessary. In operation, the balloon is typically deflated while the catheter is inserted into a body cavity (e.g., a heart) of a patient, inflated in order to perform the necessary procedure, and deflated again upon completing the procedure.

(3) U.S. Pat. No. 5,865,801 to Houser, whose disclosure is incorporated herein by reference, describes a catheter with a dilatation balloon fixed to the catheter tubing near its distal end. The catheter may include piezoelectric ultrasonic transducers on the balloon's exterior surface allowing a physician to utilize ultrasound to position the catheter and to view a surface surrounding the catheter.

(4) U.S. Pat. No. 8,819,928 to Nix et al., whose disclosure is incorporated herein by reference, describes an ultrasonic imaging catheter. The catheter includes a flexible circuit electrically coupled to a transducer array mounted on a distal end of the catheter, and a balloon at the distal end close to the transducer array. The catheter includes a stent mounted on the balloon, the stent carrying one or more drugs designed to be eluted or washed into a patient's blood stream after the stent has been inserted into a target area within the patient's vascular system.

(5) International Patent WO 88/09150 to Griffith et al., whose disclosure is incorporated herein by reference, describes an ultrasonic imaging array and balloon catheter assembly. The catheter includes an array of miniature ultrasound crystals that are mounted on a preassembled subassembly, which is, in turn, mounted on a small lumen catheter that provides dimensional and other quantitative information relating to arterial wall geometry and character at disease or obstruction sites. Balloons are also mounted on the catheter to make it possible to use the catheter for an angioplasty procedure.

(6) U.S. Pat. No. 5,722,972 to Power et al., whose disclosure is incorporated herein by reference, describes an excimer laser catheter that can be used for ablation of atherosclerotic blockage. The catheter includes a tube-like basic body with a distal end and a proximal end, at least two balloon members positioned at the distal end, and ultrasonic probe positioned between two of the balloon members. The ultrasonic probe includes a series of piezo crystals and at least one multiplexer which can activate the piezo crystals in a phased manner.

(7) International patent WO 93/04727 to McNicholas et al., whose disclosure is incorporated herein by reference, describes a balloon catheter that can be used to treat body tissue. The catheter includes a balloon and means for ultrasonically viewing prostatic tissue.

(8) U.S. Pat. No. 4,917,097 to Proudian et al., whose disclosure is incorporated herein by reference,

describes an in vivo imaging device that is configured to image small body cavities.

(9) Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that to the extent any terms are defined in these incorporated documents in a manner that conflicts with the definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

(10) The description above is presented as a general overview of related art in this field and should not be construed as an admission that any of the information it contains constitutes prior art against the present patent application.

SUMMARY OF THE INVENTION

(11) There is provided, in accordance with an embodiment of the present invention, a medical apparatus, including a probe having a distal end configured for insertion into a body cavity and containing a lumen that opens through the distal end, an inflatable balloon deployable through the lumen into the body cavity, the balloon having an exterior wall, a flexible printed circuit board having a first side attached to the exterior wall of the inflatable balloon and a second side opposite the first side, and an ultrasonic transducer mounted on the first side of the flexible printed circuit board and encapsulated between the exterior wall of the balloon and the flexible printed circuit board.

(12) In some embodiments, the ultrasonic transducer includes a piezo ceramic crystal. In additional embodiments, the ultrasonic transducer is configured to operate in amplitude mode. In further embodiments, the ultrasonic transducer is configured to generate a signal between one and ten megahertz in order to determine a distance between the ultrasonic transducer and tissue in the body cavity. In supplemental embodiments, the ultrasonic transducer can be configured to generate a signal greater than twenty megahertz in order to determine a thickness of tissue in contact with the second side of the flexible circuit board.

(13) In some embodiments, the medical apparatus may include an electrode mounted on the second side of the flexible circuit board and configured as a location sensor. In additional embodiments wherein the medical apparatus include the electrode, the medical apparatus may include a processor configured to generate a map of the body cavity based on a distance between the ultrasonic transducer and tissue in the body cavity and a signal from the electrode. In further embodiments, the body cavity includes a chamber of a heart.

(14) There is also provided, in accordance with an embodiment of the present invention, a method, including providing a probe having a distal end configured for insertion into a body cavity and containing a lumen that opens through the distal end, providing an inflatable balloon deployable through the lumen into the body cavity, the balloon having an exterior wall, attaching a first side of a flexible printed circuit board to the exterior wall of the inflatable balloon, the flexible circuit board having a second side opposite the first side, and mounting an ultrasonic transducer on the first side of the flexible printed circuit board, thereby encapsulating the ultrasonic transducer between the exterior wall of the balloon and the flexible printed circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is herein described, by way of example only, with reference to the accompanying drawings, wherein:

(2) FIG. 1 is a schematic, pictorial illustration of a medical system comprising a balloon catheter with a distal end, in accordance with an embodiment of the present invention;

(3) FIG. 2 is a schematic pictorial illustration of the distal end comprising multiple flexible printed circuit boards affixed to a balloon, in accordance with an embodiment of the present invention;

(4) FIG. 3 is a schematic exploded view showing an ultrasonic transducer mounted on a given

- flexible circuit board, in accordance with an embodiment of the present invention;
- (5) FIG. 4 is a schematic view of the distal end of the balloon catheter when the balloon is inflated, in accordance with an embodiment of the present invention;
- (6) FIGS. 5A and 5B are schematic illustrations of high frequency signals that are generated by the ultrasonic transducer, in accordance with an embodiment of the present invention;
- (7) FIG. 6 is a schematic view of the distal end of the catheter positioned within a cardiac chamber, in accordance with an embodiment of the present invention; and
- (8) FIG. 7 is a schematic view of the distal end of the catheter in contact with a wall of the cardiac chamber, in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION OF EMBODIMENTS

Overview

- (9) Balloon catheters can be configured to perform invasive procedures such as intracardiac ablation and anatomical mapping. While performing intracardiac ablation, balloon catheters can typically ablate larger areas of tissue than traditional single tip catheters that perform point-by-point ablation. Furthermore, balloon catheters can ablate the larger areas of tissue at a single time, reducing procedure time and exposure to radiation related to the use of fluoroscopy during the procedure.
- (10) During an anatomical mapping procedure, a map is created that comprises map points collected from the balloon catheter. Each map point comprises a respective coordinate within a body cavity, and while mapping a body cavity such as a chamber of a heart, the map points may be registered to a pre-acquired image of the body cavity, thereby providing a practical visualization of the body cavity.
- (11) In embodiments of the present invention, a balloon catheter comprises one or more ultrasonic transducers that can be used to map a body cavity, and to measure thicknesses of walls in the body cavity during an ablation procedure. As described hereinbelow, the balloon catheter comprises a distal end configured for insertion into a body cavity and containing a lumen that opens through the distal end. The balloon catheter also comprises an inflatable balloon deployable through the lumen into the body cavity, the balloon having an exterior wall, and a flexible printed circuit board having a first side attached to the exterior wall of the inflatable balloon and a second side opposite the first side. The balloon catheter additionally comprises an ultrasonic transducer mounted on the first side of the flexible printed circuit board and encapsulated between the exterior wall of the balloon and the flexible printed circuit board.
- (12) While inserted into a body cavity (e.g., a heart), the ultrasonic transducers can transmit and receive a high frequency signal, and analyze the signal in order to determine a current distance between the ultrasonic transducers and a wall of the body cavity. As a medical professional maneuvers the distal end within the body cavity and collects distances between the distal end and walls of the body cavity, the distances can be used to generate a map of the body cavity. Additionally, when the distal end of the balloon catheter is in contact with tissue in the body cavity (e.g., during an ablation procedure), the high frequency signal can be analyzed to determine a thickness of the tissue.
- (13) In some embodiments, each given ultrasonic transducer may comprise a piezo ceramic crystal, which is typically brittle. Encapsulating the ultrasonic transducers between the exterior wall of the balloon and the flexible printed circuit board can help protect and prevent the ultrasonic transducers from breaking when the distal end of the balloon catheter is in contact with tissue in the body cavity.

System Description

- (14) FIG. 1 is a schematic, pictorial illustration of a medical system **20** comprising a medical probe **22** and a control console **24**, and FIG. 2 is a schematic pictorial illustration of a distal end **26** of the medical probe, in accordance with an embodiment of the present invention. System **20** may be based, for example, on the CARTO® system, produced by Biosense Webster Inc. (Diamond Bar,

California, U.S.A.). In embodiments described hereinbelow, medical probe **22** comprises a balloon catheter that is used for diagnostic or therapeutic treatment, such as for mapping electrical potentials and/or for performing ablation procedures in a heart **28** of a patient **30**. Alternatively, probe **22** may be used, mutatis mutandis, for other therapeutic and/or diagnostic purposes in the heart or in other body organs.

(15) During a medical procedure, a medical professional **32** inserts medical probe **22** into a biocompatible sheath (not shown) that has been pre-positioned in a lumen of the patient so that a balloon **34** (FIG. 2) affixed to distal end **26** of the medical probe enters the lumen (e.g., a chamber of heart **28**). Balloon **34** is typically formed from bio-compatible material such as polyethylene terephthalate (PET), polyurethane, nylon, or silicone.

(16) Console **24** is connected, by a cable **36**, to body surface electrodes, which typically comprise adhesive skin patches **38** that are affixed to patient **30**. Console **24** comprises a processor **40** that determines position coordinates of distal end **26** inside heart **28** based on impedances measured between adhesive skin patches **38** and one or more electrodes **42** (also referred to as microelectrodes **42**) that are mounted on a flexible circuit board **44** that is affixed to balloon **34**. In the configuration shown in FIG. 2, distal end **26** comprises multiple flexible circuit boards **44** affixed to balloon **34**, each of the circuit boards comprising a pair of microelectrodes **42**.

(17) Processor **40** typically comprises a general-purpose computer, with suitable front end and interface circuits for receiving signals from elements of probe **22** (e.g., microelectrodes **42**) and controlling the other components of console **24**. Processor **40** may be programmed in software to carry out the functions that are described herein. The software may be downloaded to console **24** in electronic form, over a network, for example, or it may be provided on non-transitory tangible media, such as optical, magnetic or electronic memory media. Alternatively, some or all of the functions of processor **40** may be carried out by dedicated or programmable digital hardware components.

(18) Although the medical system shown in FIG. 1 uses impedance-based sensing to measure a location of distal end **26**, other position tracking techniques may be used (e.g., techniques using magnetic-based sensors). Impedance-based position tracking techniques are described, for example, in U.S. Pat. Nos. 5,983,126, 6,456,864 and 5,944,022, whose disclosures are incorporated herein by reference. Magnetic position tracking techniques are described, for example, in U.S. Pat. Nos. 5,391,199, 5,443,489, 6,788,967, 6,690,963, 5,558,091, 6,172,499, 6,177,792, whose disclosures are incorporated herein by reference. The method of position sensing described hereinabove is implemented in the above-mentioned CARTO™ system and is described in detail in the patents cited above.

(19) Console **24** also comprises an input/output (I/O) communications interface **46** that enables console **24** to transfer signals from, and/or transfer signals to electrodes **42** in probe **22** and adhesive skin patches **38**. In embodiments described hereinbelow, processor **40** can use signals received from microelectrodes **42** and adhesive skin patches **38** to generate a map **48** that shows the position of balloon **34** in the patient's body. During the procedure, processor **40** can present map **48** to medical professional **32** on a display **50**, and store data representing the map in a memory **52**. Memory **52** may comprise any suitable volatile and/or non-volatile memory, such as random access memory or a hard disk drive. In some embodiments, medical professional **32** can manipulate map **48** using one or more input devices **54**.

(20) As shown in FIG. 2, distal end **26** comprises multiple flexible circuit boards **44** mounted on balloon **34**, each of the flexible circuit boards comprising a pair of microelectrodes **42**, and an electrode **43** (also referred to herein as an ablation electrode **43**) that can be used to ablate cardiac tissue in heart **28**. Flexible circuit board **44** comprises an insulated substrate, typically implemented with polyimide, and electrodes **42** and **43** typically comprise gold overlaying the substrate.

(21) Control console **24** also comprises an ablation module **56** and a pulser/receiver circuit **58**. Control console **24** is configured to monitor and control ablation parameters such as the level and

the duration of ablation power applied to the ablation electrodes. Pulser/receiver circuit **58** may be based, for example, on JSR Ultrasonics' DPR-300 Pulser/Receiver, produced by Imaginant (Pittsford, NY, U.S.A.), and the functionality of pulser/receiver circuit **58** is described hereinbelow. (22) FIG. **3** is a schematic side view of a portion of distal end **26**, showing components of the distal end, in accordance with an embodiment of the present invention. Balloon **34** comprises an exterior wall **60**, and flexible circuit board **44** has a first side and a second side **64** opposite the first side. Each given microelectrode **42** is coupled, by a conductor **72**, to circuitry (including processor **40**) in control console **24** that enables processor **40** to determine a location of the given microelectrode, and each ablation electrode **43** is coupled to ablation module **56** by a conductor **74**.

(23) In embodiments of the present invention, an ultrasonic transducer **68** is mounted on first side **62** of the flexible circuit board, electrodes **42** and **43** are mounted on second side **64**, and the first side is then attached to exterior wall **60**, as indicated by arrows **66**. Ultrasonic transducer **68** is coupled to pulser/receiver circuit **58** by a connection **70**, and may comprise a piezo ceramic crystal, such as lead zirconate titanate (PZT), lead titanate (PT), or lead metaniobate (PbNb.sub.2O.sub.6).

(24) In operation, medical system **20** typically operates in amplitude mode (also known as A-mode). While operating in A-mode, pulser/receiver circuit **58** conveys a first electrical signal to ultrasonic transducer that excites the piezo ceramic crystal, and as a result of exciting the piezo ceramic crystal, ultrasonic transducer **68** transmits a unidirectional high frequency signal that travels as a relatively narrow beam.

(25) While distal end **26** is positioned into a body cavity such as heart **28**, the high frequency signal will bounce (i.e., reflect) off of surfaces in patient **30**. Upon returning to ultrasonic transducer **68**, the reflected signal deforms the piezo ceramic crystal, causing the piezo ceramic crystal to generate a second electrical signal that is conveyed to pulser/receiver circuit **58**. Processor **40** can then measure the time between the first and the second electrical signals in order to determine a distance between ultrasonic transducer **68** and the respective surfaces. As described in the description referencing FIG. **5** hereinbelow, surfaces that can reflect the high frequency signal when distal end is inserted into heart **28** may include second side **64** and tissue in the body cavity.

(26) In the configuration shown in FIG. **3**, electrodes **42** and **43** are mounted on second side **64** of flexible circuit board **44**. While embodiments described herein have microelectrodes **42** configured as location sensors, configuring the microelectrodes to perform other tasks during a medical procedure is considered to be within the spirit and scope of the present invention. For example, some or all of microelectrodes **42** may be configured to measure electrical activity of heart **28**. Additionally or alternatively, ablation electrodes **43** may also be configured as location sensors (i.e., in addition to or instead of being configured to ablate tissue).

(27) In some embodiments, flexible circuit board **44** may comprise a thermocouple **76** mounted on first side **62**. Thermocouple **76** may be coupled to ablation electrode **43** by a conductive via **78**, and the thermocouple may be formed as a junction between a copper conductor **80** and a constantan conductor **82**. Conductors **80** and **82** are typically connected to ablation module **56** and enable the module to monitor the temperature of ablation electrode **43** during an ablation procedure. Conductors **70**, **72**, **74**, **80** and **82** can be formed as conducting lines embedded in the substrate of flexible circuit board **44**.

(28) FIG. **4** is a schematic side view of distal end **26** when balloon **34** is inflated, in accordance with an embodiment of the present invention. Balloon **34** is affixed to a tubular shaft **92** that is configured to extend from a lumen **90** at a distal end of medical probe **22**, and balloon **34** is configured to be deployed through the lumen into a body cavity such as heart **28**. As shown in FIG. **4**, and thermocouple **76** are ultrasonic transducer **68** encapsulated between exterior wall **60** of the balloon and first side **62** (see first side **62** and transducer **68** in FIG. **3**) of flexible printed circuit board **44**.

(29) FIGS. **5A** and **5B**, referred to collectively as FIG. **5**, are schematic illustrations of high frequency signals that are generated by ultrasonic transducer **68**, in accordance with an

embodiment of the present invention. Pulser/receiver circuit **58** conveys a high frequency electrical pulse to transducer **68**, which then transmits a high frequency signal of pulsed ultrasound from transducer **68** when it operates in the A-mode. As explained below the high frequency ultrasound signal travels along a signal path, and the signal separates into a number of different signals of pulsed ultrasound having the same frequency but reduced amplitude compared with the original signal. The reduced amplitude signals travel along respective associated signal paths.

(30) In the example shown in FIG. 5, the different signals and their associated signal paths are generically termed signals and paths **100**. The signals and their associated paths are differentiated herein by appending a letter to the identifying numeral, so that different signals have associated signal paths **100A-100D**.

(31) Thus, as shown in FIG. 5A, on receipt of the electrical pulse from circuit **58**, ultrasonic transducer **68** transmits an initial high frequency ultrasound signal, and the signal travels through flexible circuit board **44** along a signal path **100A** until it strikes an interface formed by electrode **43** contacting second side **64**. At the interface, i.e. at second side **64**, a portion of the initial signal reflects as a first high frequency signal, which travels back to the ultrasonic transducer along a path **100B**.

(32) As shown in FIG. 5B, the remainder of signal transmits through the interface at side **64**, as a high frequency signal which travels along a path **100C**. Signal continues traveling along path **100C** until it strikes the surface of endocardial tissue **102** in heart **28**, at which point it reflects as a high frequency signal which travels along a path **100D** back to the ultrasonic transducer.

(33) While for clarity the paths of the signals have been separated in FIG. 5, it will be understood that the paths may actually be substantially collinear, for example in the case that the paths are normal to transducer **68** and side **64**. Nonetheless, the paths are separate in time.

(34) From a knowledge of the times of transmittal of signal from transducer **68**, and of receipt of signal at the transducer, pulser/receiver circuit **58** is able to evaluate the “time of flight” of the ultrasound signal generated by the transducer, and provides the value of the time of flight to processor **40**. Processor **40** uses the time of flight, and known values of the speed of sound in the media in which the ultrasound pulse travels, as well as the thicknesses of board **44** and electrode **43**, to evaluate the distance between transducer **68** and tissue **102**. As described herein, the processor may use the distances from different transducers **68** to tissue **102** to generate a map of the body cavity containing distal end **26**.

(35) FIGS. 6 and 7 are schematic detail views of distal end **26** in a chamber of heart **28**, in accordance with an embodiment of the present invention. In FIG. 6, balloon **34** is inflated and deployed through lumen **90** (see FIG. 4) but does not contact endocardial tissue **102**, and in FIG. 7, the balloon is pressed against endocardial tissue **102** in heart **28** so that one or more electrodes **43** are in contact with the endocardial tissue.

(36) As shown in FIG. 6, while balloon **34** is positioned in a body cavity such as a chamber of heart **28**, but not in contact with endocardial tissue **102**, processor **40** can use location measurements from microelectrodes **42** to generate a three-dimensional software model that represents balloon **34** and locations of ultrasonic transducers **68**. As medical professional **32** maneuvers distal end **26** within the cardiac chamber, ultrasonic transducers **68** can transmit and receive first high frequency signals (typically at frequencies between 1-10 megahertz), and generated in response to corresponding electrical signals from pulser/receiver circuit **58**) that enable processor **40** to determine distances **110** between ultrasonic transducers **68** and endocardial tissue **102**. Processor **40** can use the locations of ultrasonic transducers **68** (i.e., determined by the three-dimensional model) and distances **110** to construct map **48**.

(37) In FIG. 7, ablation electrodes **43** are in contact with endocardial tissue **102** (e.g., while the ablation electrodes are performing an ablation procedure). In this situation circuit **58** is configured to convey second high frequency electrical signals (typically at frequencies greater than 20 megahertz) to ultrasonic transducers **68**, which in turn transmit and receive second high frequency

ultrasound signals that enable processor **40** to determine distances **120** between the ultrasonic transducers and epicardial tissue **122**. Distances **120** indicate thicknesses at various locations on the wall of heart **28**.

(38) In operation, the second higher frequency signals used to determine distances **120** have a higher resolution and a shorter range than the first high frequency signals used to determine distances **110**. Additionally, processor **40** can be configured to use different calculation factors when determining distances **110** and **120**, since the high frequency signals typically travel at different speeds in different media (i.e., blood in FIG. **6**, and tissue in FIG. **7**).

(39) It will be appreciated that the embodiments described above are cited by way of example, and that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

Claims

1. A medical apparatus, comprising: a probe having a distal end configured for insertion into a body cavity of an organ and containing a lumen that opens through the distal end; an inflatable balloon deployable through the lumen into the body cavity, the balloon having an exterior wall; a flexible printed circuit board having a first side attached to the exterior wall of the inflatable balloon and a second side opposite the first side, the second side spaced apart from the first side, the flexible printed circuit board having an ablation electrode and a pair of micro-electrodes disposed on the second side of the flexible printed circuit board and configured to measure electrical activity of the organ; (1) a thermocouple coupled to the ablation electrode by a conductive via; and (2) an ultrasonic transducer mounted on the first side of the flexible printed circuit board and disposed between the exterior wall of the balloon and the flexible printed circuit board; wherein when the balloon is deployed into the body cavity the micro-electrodes provide location signals of the balloon location and the ultrasonic transducer location.
2. The medical apparatus according to claim 1, wherein the ultrasonic transducer comprises a piezo ceramic crystal.
3. The medical apparatus according to claim 1, wherein the ultrasonic transducer is configured to operate in amplitude mode.
4. The medical apparatus according to claim 1, wherein the ultrasonic transducer is configured to generate a signal between one and ten megahertz in order to determine a distance between the ultrasonic transducer and tissue in the body cavity.
5. The medical apparatus according to claim 1, and further comprising a processor configured to generate a map of the body cavity based on a distance between the ultrasonic transducer and tissue in the body cavity and the location signals from the micro-electrodes.
6. The medical apparatus according to claim 5, wherein the body cavity comprises a chamber of a heart.
7. A method, comprising: providing a probe having a distal end configured for insertion into a heart and containing a lumen that opens through the distal end; providing an inflatable balloon deployable through the lumen into the body cavity, the balloon having an exterior wall; attaching a first side of a flexible printed circuit board to the exterior wall of the inflatable balloon, the flexible circuit board having a second side opposite the first side, the second side spaced apart from the first side, the flexible printed circuit board having an ablation electrode and a pair of micro-electrodes on the second side; coupling a thermocouple mounted on the first side of the flexible printed circuit board to the ablation electrode mounted on the second side by a conductive via, thereby disposing the thermocouple between the exterior wall of the balloon and the flexible printed circuit board;

and mounting an ultrasonic transducer on the first side of the flexible printed circuit board, thereby disposing the ultrasonic transducer between the exterior wall of the balloon and the flexible printed circuit board; wherein when the balloon is deployed into the body cavity the micro-electrodes provide location signals of the balloon location and the ultrasonic transducer location.

8. The method according to claim 7, wherein the ultrasonic transducer comprises a piezo ceramic crystal.

9. The method according to claim 7 and comprising operating the ultrasonic transducer in amplitude mode.

10. The method according to claim 7, and comprising generating, by the ultrasonic transducer, a signal between one and ten megahertz, and determining, by a processor based on the signal, a distance between the ultrasonic transducer and tissue in the body cavity.

11. The medical apparatus according to claim 10, and further comprising generating, by the processor, a map of the body cavity based on the distance between the ultrasonic transducer and tissue in the heart and the location signals from the pair of micro-electrodes.

12. The method according to claim 7, and comprising generating, by the ultrasonic transducer, a signal, and determining by a processor based on the signal, a thickness of tissue in contact with the second side of the flexible circuit board.
